Experiment -2

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Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

FRAME 1:
INSTRUCTION PAGE:

Chunking Analysis of the Instruction Page

Chunking is a cognitive strategy that breaks down information into smaller, manageable units, making it easier to process and retain. The **Memory Recall Task** instruction page effectively utilizes chunking in the following ways:

1. Clear and Sequential Numbering

· The instructions are **broken down into four steps**, making it easier to follow. · Each step presents **one key action** in a structured manner, reducing cognitive overload.

2. Logical Grouping of Information

· **Observation Phase (Step 1)** → Users learn about what they will see.

· **Memorization Strategy (Step 2)** → Encourages users to focus on remembering items.

· **Recall Phase (Step 3)** → Explains how users will recall information.

· **Key Performance Metric (Step 4)** → Highlights that accuracy matters more than speed.

3. Visual Hierarchy and Design Elements

· **Large title ("MEMORY RECALL TASK")** → Grabs attention and clearly states the task. · **Bullet points and spacing** → Reduce clutter, enhancing readability.

· **Highlighted "START" button** → Signals the next step, keeping navigation intuitive.

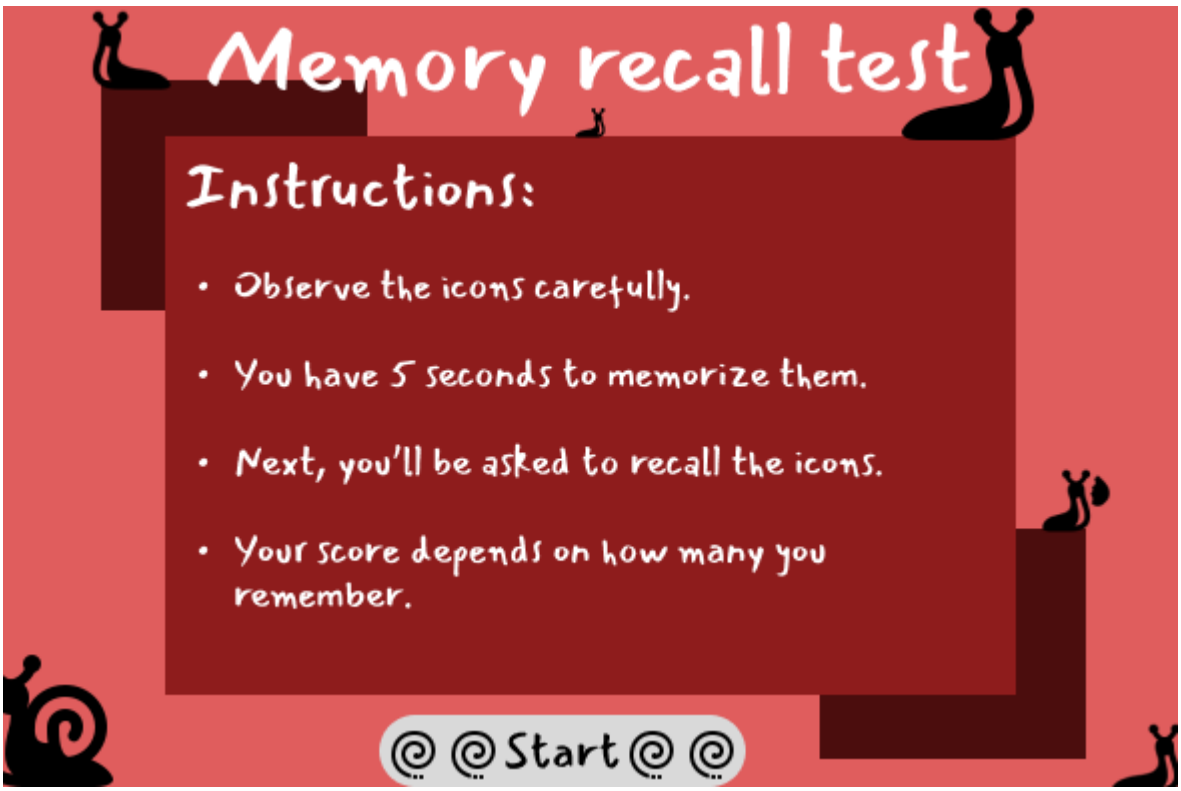
· . **Insect and Emojis visuals** → Add a playful touch, engaging users without distraction.

4. Time Constraint Reinforcement

- Step 2 explicitly states, **"You have 5 seconds to memorize them."**
- This **reinforces expectations** while subtly urging users to focus.

5. Simplicity and Clarity

- Instructions use **short, direct sentences**, avoiding unnecessary complexity.



FRAME 2:

CHUNKING PHASE:

Analysis of the Memory Recall Task - Chunking Phase Screen

This screen represents the **Chunking Phase** of a **Memory Recall Task**, where users observe and memorize different icons within a limited time. Below is a breakdown of its key components:

1. Purpose of the Screen

- This is the **visual memory encoding phase**, where users **view and group items mentally** before recalling them.

- The term **"Chunking Phase"** suggests that the game applies the **chunking principle**, which helps users remember items by categorizing them into meaningful groups.

2. Key Elements and UI Components

☒ **Countdown Timer (Top Left - Red Circle: "00:05")**

- Indicates that users have **5 seconds** to observe and memorize the displayed items.

- The bright **red color** and bold text create urgency, ensuring users stay focused.

Time is ticking…(Text below the title)

- Reinforces the sense of urgency.

Grid of Icons

- Various **icons** (such as a monster, spy, dollar, leaf, monkey, fountain, tacos, etc.) are displayed in a **5x4 grid format**.
- These icons are **visually distinct yet grouped by similarities**, encouraging **chunking strategies** like:
  - **Categorizing by theme** (e.g., animals, food,leaf).
  - **Grouping similar colors or backgrounds** (e.g., pink, blue, or yellow tiles).
  - **Associating repeated icons** (e.g., multiple tacos, spies, monkeys).

Hourglass

. Beside the “Time is ticking..”, indicated the seconds are running by, urging the users to remember icons as much as possible.

3. How the Chunking Phase Works

1. Users **scan the grid** and look for **patterns or related items** to create mental chunks.
2. **The countdown timer limits observation time**, forcing quick memory strategies.
3. Once time is up, users **transition to the recall phase**, where they must identify previously seen items.

4. Cognitive and UX Benefits of Chunking

- **Enhances short-term memory** by allowing users to recall **groups of information** instead of individual elements.
- **Reduces cognitive overload** by helping users organize data efficiently.
- **Improves pattern recognition**, making recall easier and more accurate.

FRAME 3:

RECALL PHASE:

Analysis of the Memory Recall Task - Selection Phase

This screen represents the **Selection Phase** of a **Memory Recall Task**, where users recall and choose the items they remember from the previous **Chunking Phase**. Below is a breakdown of its components:

1. Purpose of the Screen

- This is the **memory retrieval** stage, where users **select the items they remember seeing** in the previous phase.
- The goal is to **test the effectiveness of chunking and short-term memory retention**.

2. Key Elements and UI Components

Title & Instructions

- **"SELECT THE ICONS THAT YOU THINK YOU SAW"**– Clear instruction guiding the user to choose remembered items.

Grid of Icon Choices

- A set of **10 icons** are presented as multiple-choice options.
- Some icons were previously displayed, while others are **distractors (new icons added to confuse users).**
- Users must identify which icons appeared in the **Chunking Phase**.

"SUBMIT" Button (Bottom Center)

- Once users have made their selections, they press **"SUBMIT"** to confirm their recall choices.

3. How the Selection Phase Works

1. Users **analyze the displayed icons** and recall which ones they saw in the **Chunking Phase**.
2. They **select the remembered icons.**
3. Some **icons are distractors**, testing whether the user's memory is accurate or if they mistakenly recall incorrect icons.
4. Clicking **"SUBMIT"** finalizes their choices, leading to a results or feedback screen.

4. Cognitive and UX Benefits

- **Tests memory accuracy** by comparing user selections with previously displayed items. ·

. . **Incorporates distractors** to evaluate how well users distinguish real vs. false memories.

- . . **User-friendly interface** with clear selection mechanics.

- **Gamified elements** (bright colors) make the task engaging.

FRAME 4:
RESULT PAGE:

Analysis of the Memory Recall Task - Score Screen

This screen represents the **Score** of the **Memory Recall Task**, where users receive their performance evaluation based on the selections made in the previous **Recall Phase**.

1. Purpose of the Screen

- Provides **feedback on recall accuracy** by showing the number of correct answers.
- Allows users to try again.

2. Key Elements & UI Components

Title & Score Display

- "**SCORE"** – Indicates that the user's performance is being displayed.
- A **digital scoreboard** with the user's score **(1/5)** in bold, showing correct answers out of total attempts.

. A "Really?" text with a disappointing emoji is on the top of the score board expressing the disappointment of the score that has been scored by the user.

Action Buttons

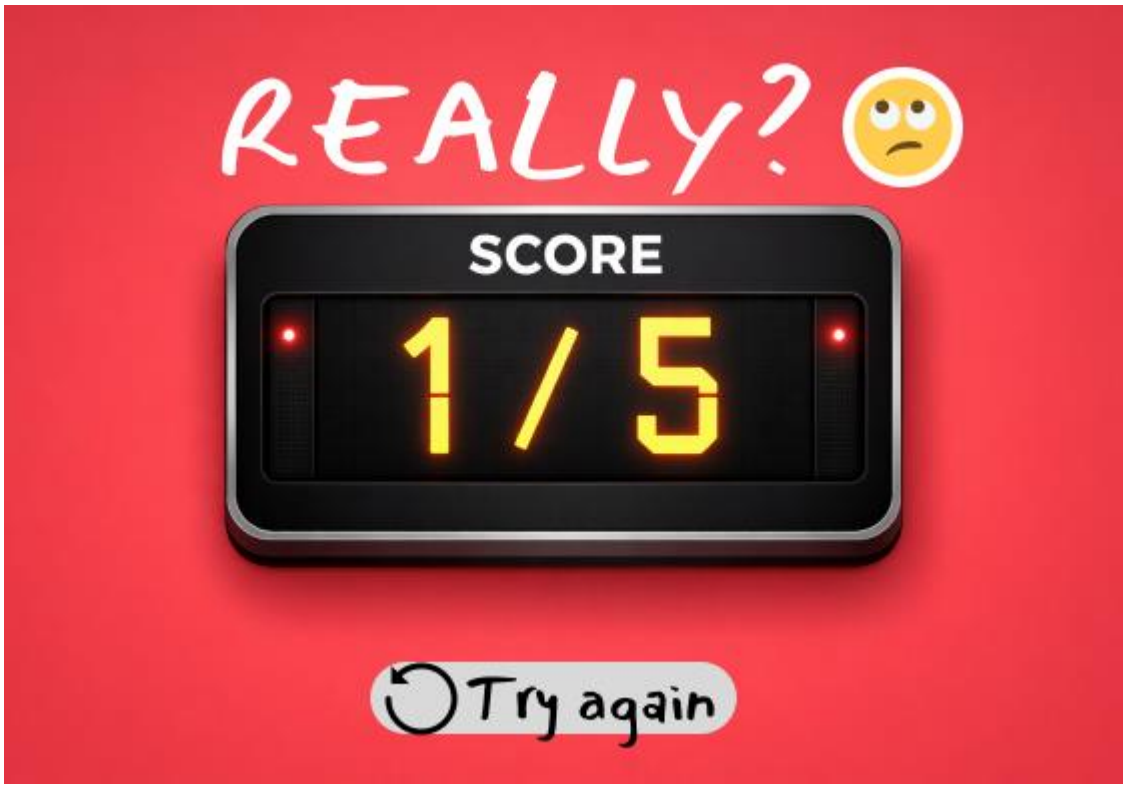
. “Try again” with a replay icon allows the user to have an another attempt.

3. How This Phase Works

1. The game evaluates the user’s selections from the **Recall Phase**.
2. It calculates the **accuracy score (1/5 in this case)** and displays it on a **digital board**.
3. Users **review their performance** and choose their next action:
   - **Try again**if they want to improve their recall.

4. Cognitive & UX Benefits

- **Instant feedback** helps users track their memory performance.
- **Another attempt (Try again)** gives users control over their learning experience.
- **Visual & gamified elements** make the task engaging and less stressful.



PROTOTYPE LINK:

https://www.figma.com/proto/N3jDwSVQb8xekoFpx74oKE/Untitled?node-id=21-1925&t=wgeq5IQtJfxYzsN3-0&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=4%3A62